Bond Risk, ML and Monte Carlo RL

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EXPERIMENTAL RESEARCH OUTPUT. Historical portfolio returns are an approximate rate-only book proxy, not exact historical portfolio P&L. Credit spreads, MBS optionality, FX and liquidity risks are omitted from that proxy. Scenario repricing is a separate calculation. RL controls exposure; it does not implement an instrument-level hedge.

Findings

ML held-out RMSE is 0.001208; trailing-volatility RMSE is 0.001280. ML error is lower.

RL seed summaries are descriptive averages and standard deviations across training seeds, not confidence intervals.

Reduced exposure can lower drawdown without demonstrating a superior policy; compare return, downside risk and turnover.

Method limitations

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ML predicts next-five-day daily RMS returns, not expected returns.

The ML-scaled Monte Carlo model is an experimental adjustment.

The five-day volatility forecast is held constant for 20-day paths.

Monte Carlo distributions are descriptive, not coverage backtests.

RL evaluation uses held-out history and simplified transaction costs.

Multiple RL seeds are shown; no best seed is selected for reporting.

No guarantee of predictive skill or profitable deployment is implied.

Portfolio

index	sleeve	weight	notional_usd	price	mod_dur	convexity
UST_2Y	treasuries	0.06	6e+08	98.72	1.892	4.6
UST_5Y	treasuries	0.1	1e+09	96.76	4.425	23
UST_10Y	treasuries	0.14	1.4e+09	96.21	7.806	73.7
UST_30Y	treasuries	0.1	1e+09	92.18	14.673	325.8
IG_7Y	ig_corp	0.3	3e+09	100.06	5.806	40.1
HY_4Y	hy	0.1	1e+09	108.81	3.465	14.6
MBS_5Y5	mbs	0.1	1e+09	101.36	4.716	26.4
CASH	cash	0.1	1e+09		0	0
TOTAL	TOTAL	1	1e+10			

index	dv01_usd	krd_2y_usd	krd_5y_usd	krd_10y_usd	krd_30y_usd
UST_2Y	112,092	113,774	-0	-0	-0
UST_5Y	428,203	-3,979.48	447,724	-0	-0
UST_10Y	1.05139e+06	-2,868.86	-11,000.9	1.11309e+06	-0
UST_30Y	1.35262e+06	-784.492	-3,008.21	-34,311	1.51409e+06
IG_7Y	1.74276e+06	140.92	1.04926e+06	699,454	-0
HY_4Y	376,992	121,568	225,667	-0	-0
MBS_5Y5	477,989	1,426.66	424,546	46,993.1	-0
CASH	0	0	0	0	0
TOTAL	5.54205e+06	229,277	2.13319e+06	1.82523e+06	1.51409e+06

Scenarios

scenario	rates_$M	spread_$M	haircut_$M	total_$M	total_%NAV	dur_only_$M
japan_baseline	-263.316	-52.0231	-30	-345.339	-3.45339	-269.371
japan_severe	-611.231	-156.069	-90	-857.3	-8.573	-652.36
japan_extreme	-1,173.4	-312.139	-180	-1,665.54	-16.6554	-1,339.52

scenario	taylor2_$M
japan_baseline	-259.743
japan_severe	-594.844
japan_extreme	-1,104.08

Historical risk

metric	value_pct
Latest 95% VaR	0.361793
Latest 95% ES	0.52002
Latest 99% VaR	0.672618
Latest 99% ES	0.746417
Worst day	-1.57891
Std dev	0.305764

VaR backtest

alpha	n_obs	expected_breaches	actual_breaches	breach_rate	expected_rate
0.95	5,421	271.05	304	0.0560782	0.05
0.99	5,421	54.21	70	0.0129127	0.01

Worst days

date	book_return
2022-06-13	-0.0157891
2008-09-19	-0.0157268
2020-03-17	-0.0140074
2008-09-30	-0.0137882
2008-03-24	-0.0130754
2020-03-10	-0.0122222
2009-06-01	-0.0119819
2010-12-07	-0.011196
2013-07-05	-0.0111045
2024-04-10	-0.0109933

Scenario history 1d

scenario	loss_pct	days_worse_or_equal	share_of_history_pct	approx_return_period_days	approx_return_period_years
japan_baseline	3.45339	0	0	inf	inf
japan_severe	8.573	0	0	inf	inf
japan_extreme	16.6554	0	0	inf	inf

Scenario history 5d

scenario	loss_pct	days_worse_or_equal	share_of_history_pct	approx_return_period_days	approx_return_period_years
japan_baseline	3.45339	0	0	inf	inf
japan_severe	8.573	0	0	inf	inf
japan_extreme	16.6554	0	0	inf	inf

ML metrics

model	MAE	RMSE
Trailing_20d	0.00095108	0.00127965
RandomForest	0.000913975	0.00120825

ML importance

index	importance
rms_60	0.428611
rms_20	0.390689
rms_5	0.10124
slope	0.0288656
yield_10y	0.0266316
mean_5	0.016951
return_today	0.00319518
return_lag1	0.00255216
dy_10y	0.00126457

Monte Carlo

method	horizon_days	confidence	VaR_pct	ES_pct	prob_loss_gt_3pct	mean_return_pct
historical_bootstrap	5	0.95	1.00444	1.35036	0	0.0440135
historical_bootstrap	5	0.99	1.59188	1.88553	0	0.0440135
ML_scaled_experiment	5	0.95	0.811917	1.0945	0	0.0446538
ML_scaled_experiment	5	0.99	1.29161	1.53231	0	0.0446538
historical_bootstrap	20	0.95	1.89664	2.45268	0.0066	0.202209
historical_bootstrap	20	0.99	2.8249	3.25542	0.0066	0.202209
ML_scaled_experiment	20	0.95	1.51449	1.97123	0.0014	0.199997
ML_scaled_experiment	20	0.99	2.27667	2.62996	0.0014	0.199997

method	paths	asof	ML_RMS_scale
historical_bootstrap	10000	2026-09-23	1
historical_bootstrap	10000	2026-09-23	1
ML_scaled_experiment	10000	2026-09-23	0.816418
ML_scaled_experiment	10000	2026-09-23	0.816418
historical_bootstrap	10000	2026-09-23	1
historical_bootstrap	10000	2026-09-23	1
ML_scaled_experiment	10000	2026-09-23	0.816418
ML_scaled_experiment	10000	2026-09-23	0.816418

RL metrics

strategy	total_return_pct	annualized_return_pct	max_drawdown_pct	daily_VaR99_pct	daily_ES99_pct	average_exposure
MC_RL	-0.169796	-0.0254424	-13.0782	0.752915	0.859697	0.756684
Full_book	-5.2497	-0.804184	-16.8933	0.81088	0.933736	1
Fixed_75pct	1.19826	0.178511	-12.613	0.603947	0.697058	0.75
Vol_target	-2.35228	-0.355787	-13.6458	0.627417	0.757974	0.83675
MC_RL	1.42827	0.212572	-12.7429	0.763228	0.863195	0.693405
Full_book	-5.2497	-0.804184	-16.8933	0.81088	0.933736	1
Fixed_75pct	1.19826	0.178511	-12.613	0.603947	0.697058	0.75
Vol_target	-2.35228	-0.355787	-13.6458	0.627417	0.757974	0.83675
MC_RL	-0.02578	-0.00386053	-13.0602	0.75108	0.858454	0.751188
Full_book	-5.2497	-0.804184	-16.8933	0.81088	0.933736	1
Fixed_75pct	1.19826	0.178511	-12.613	0.603947	0.697058	0.75
Vol_target	-2.35228	-0.355787	-13.6458	0.627417	0.757974	0.83675

strategy	total_turnover	seed
MC_RL	87.75	7
Full_book	0	7
Fixed_75pct	0.25	7
Vol_target	20.5	7
MC_RL	51	42
Full_book	0	42
Fixed_75pct	0.25	42
Vol_target	20.5	42
MC_RL	164	123
Full_book	0	123
Fixed_75pct	0.25	123

strategy	total_turnover	seed
Vol_target	20.5	123

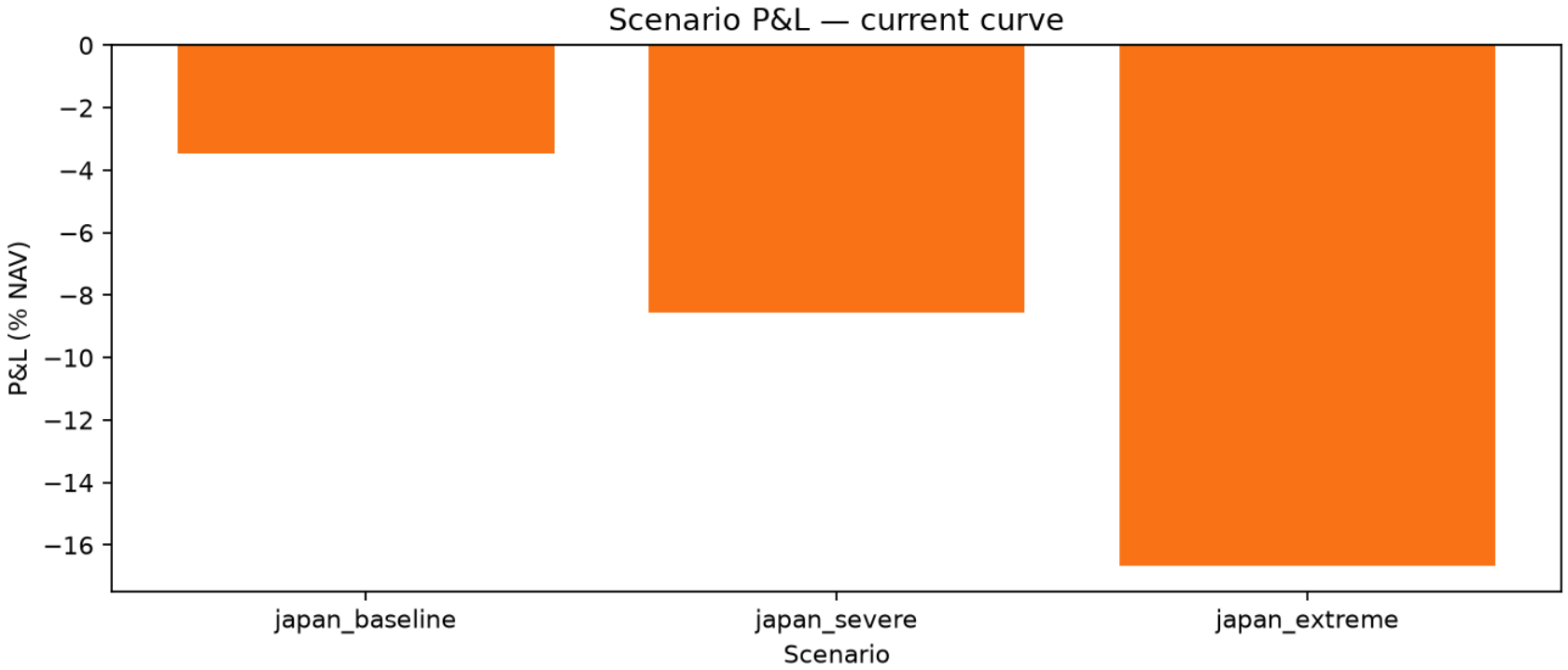
RL seed summary

strategy	total_return_pct_mean	total_return_pct_std	annualized_return_pct_mean	annualized_return_pct_std	max_drawdown_pct_mean	max_drawdown_pct_std
Fixed_75pct	1.19826	0	0.178511	0	-12.613	0
Full_book	-5.2497	0	-0.804184	0	-16.8933	0
MC_RL	0.410899	0.884009	0.0610896	0.13163	-12.9604	0.188554
Vol_target	-2.35228	0	-0.355787	0	-13.6458	0

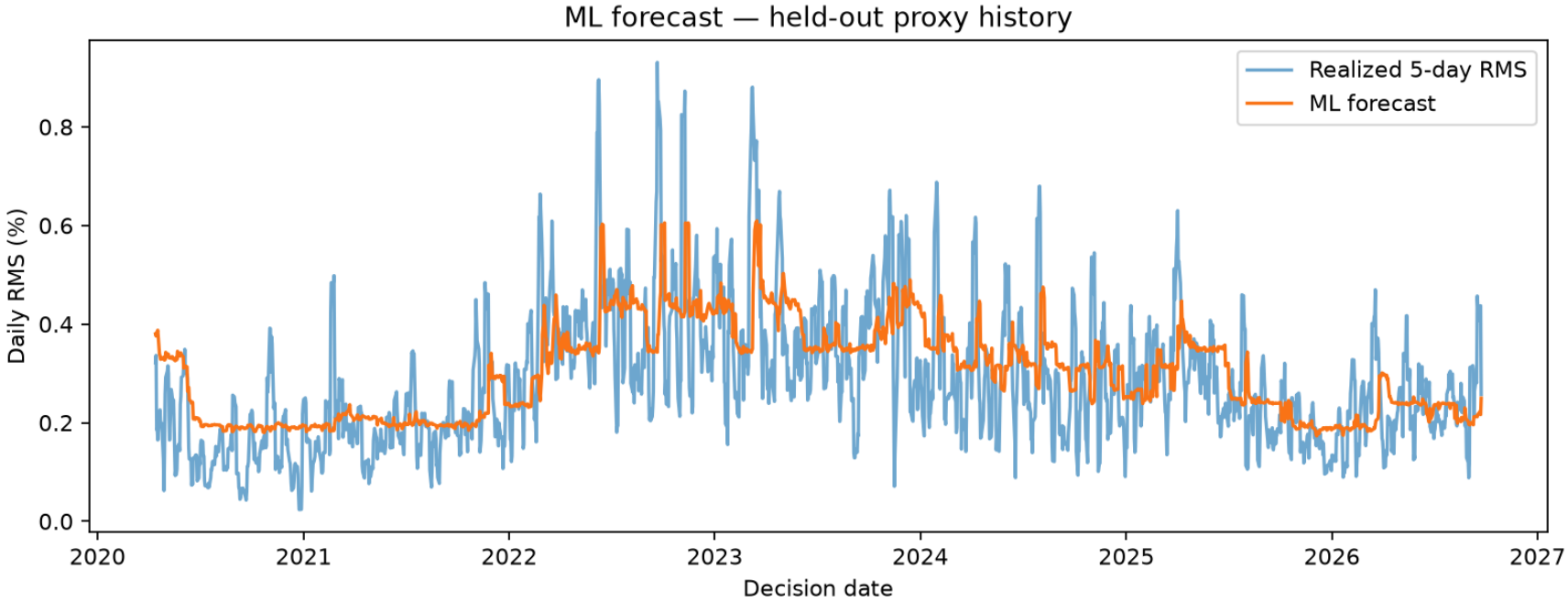
strategy	daily_VaR99_pct_mean	daily_VaR99_pct_std	daily_ES99_pct_mean	daily_ES99_pct_std	average_exposure_mean	average_exposure_std
Fixed_75pct	0.603947	0	0.697058	0	0.75	0
Full_book	0.81088	0	0.933736	0	1	0
MC_RL	0.755741	0.00654826	0.860449	0.00245827	0.733759	0.0350559
Vol_target	0.627417	0	0.757974	0	0.83675	0

strategy	total_turnover_mean	total_turnover_std
Fixed_75pct	0.25	0
Full_book	0	0
MC_RL	100.917	57.6391
Vol_target	20.5	0

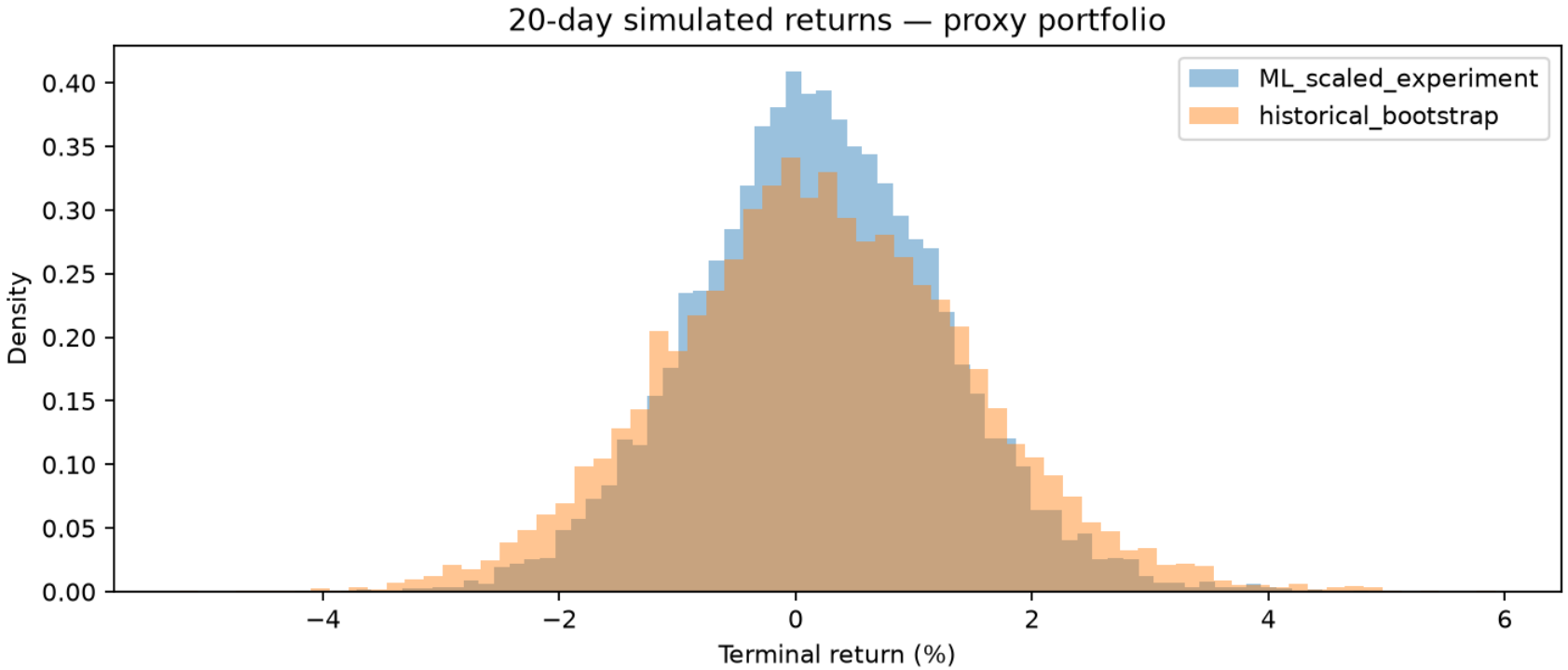
Scenario Pnl



Ml Forecast


ML forecast — held-out proxy history
Realized 5-day RMS
ML forecast
Daily RMS (%)
0.0
0.2
0.4
0.6
0.8
2020
2021
2022
2023
2024
2025
2026
2027
Decision date

Mc Distribution



Policy Equity

